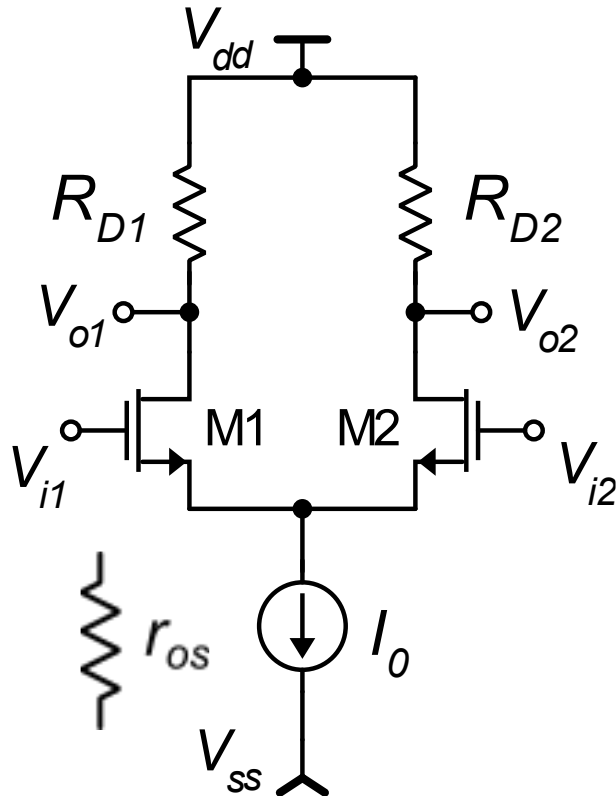


Example 1 – CMRR of Differential Amplifiers

MOS differential pair operated at a bias current of 0.8 mA employs transistors with $W/L=100$ and $\mu_n C_{ox} = 0.2 \text{ mA/V}_2$, using $R_D = 5 \text{ k}$ and $r_{os} = 25 \text{ k}$. Find the differential gain, the common-mode gain when the drain resistances have a 1% mismatch, and the CMRR.



$$g_m = \sqrt{0.2 \cdot 100 \cdot 0.8} = 4$$

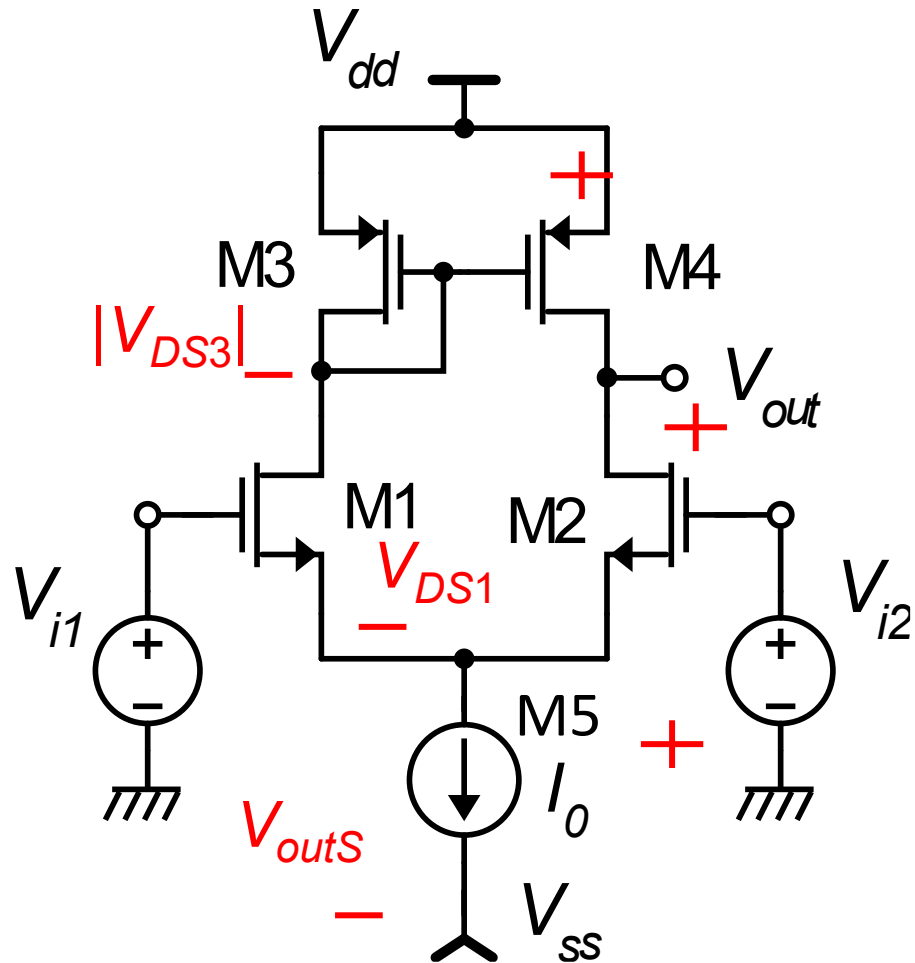
$$A_{dd} = g_m \cdot R_D = 20 \text{ V/V}$$

$$A_{cd} = \left. \frac{v_{od}}{v_{ic}} \right|_{v_{id}=0} = -\frac{1}{2r_{os}} (R_{D1} - R_{D2}) = 0.001 \text{ V/V}$$

$$CMRR = \left| \frac{A_{dd}}{A_{cd}} \right| = \frac{g_m}{2} (R_{D1} + R_{D2}) \frac{2r_{os}}{(R_{D1} - R_{D2})} = 20000$$

$$CMRR = 86 \text{ dB}$$

Example 2 – Input Common Mode Range (ICMR)



Highest Common Mode Voltage

Path from G1 through M_1 and M_3 to V_{DD} :

$$V_{IC}(\max) = V_{G1}(\max) = V_{G2}(\max) \\ = V_{DD} - V_{SG3} - V_{DS1}(\text{sat}) + V_{GS1}$$

or

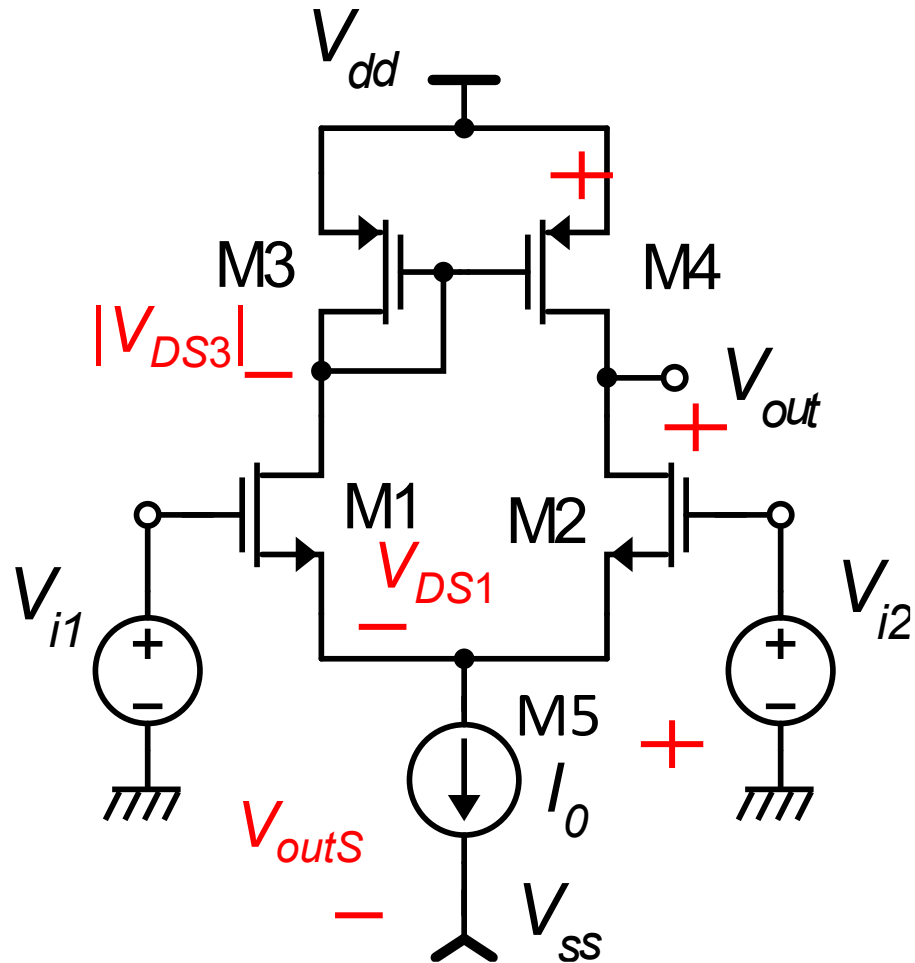
$$V_{IC}(\max) = V_{DD} - V_{SG3} + V_{TH1}$$

Path from G2 through M_2 and M_4 to V_{DD} :

$$V_{IC}(\max)' = V_{DD} - V_{SD4}(\text{sat}) - V_{DS2}(\text{sat}) + V_{GS2} \\ = V_{DD} - V_{SD4}(\text{sat}) + V_{TH2}$$

$$V_{IC}(\max) = V_{DD} - V_{SG3} + V_{TH1}$$

Example 2 – Input Common Mode Range (ICMR)



Lowest Common Mode Voltage

$$V_{IC}(\min) = V_{SS} + V_{DS5}(\text{sat}) + V_{GS1} = V_{SS} + V_{DS5}(\text{sat}) + V_{GS2}$$

where we have assumed that $V_{GS1} = V_{GS2}$ during changes in the input common mode voltage.